Inner-Product Geometry and Spectral Decomposition: From QR to Quadratic Forms

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§1.1 Diagonalization and Invariant Lines

A matrix $A \in k^{n\times n}$ is diagonalizable when there is an invertible matrix P such that

$$P^{-1}AP = D$$

is diagonal. If the columns of P are $v_1, \dots, v_n$, then

$$Av_i = \lambda_i v_i.$$

Thus diagonalization is the decomposition

$$k^n = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_s}$$

into eigenspaces. The question is not merely whether eigenvalues exist, but whether their eigenspaces supply enough independent directions.

Several properties that are often confused are logically independent. The zero matrix is diagonalizable but not invertible. The Jordan block

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is invertible but not diagonalizable. Repeated eigenvalues cause no difficulty for the identity matrix, whereas the same repeated eigenvalue leaves J with only one eigendirection. Rank, invertibility, eigenvalue multiplicity, and diagonalizability answer different questions.

§1.2 Minimal Polynomials and Eigenspaces

The minimal polynomial $m_A(t)$ is the monic polynomial of least degree satisfying

$$m_A(A) = 0.$$

If $A = PDP^{-1}$, then m_A must vanish on every diagonal entry of D, so it is a product of distinct linear factors:

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i).$$

Conversely, suppose

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i)$$

with distinct roots. Define the Lagrange polynomials

$$\ell_i(t) = \prod_{j\neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

They satisfy

$$\sum_i \ell_i(t) = 1 \pmod{m_A(t)}.$$

Hence

$$I = \sum_i \ell_i(A).$$

Moreover,

$$(A - \lambda_i I)\ell_i(A) = 0,$$

so $\ell_i(A)V \subseteq E_{\lambda_i}$. Every vector is therefore a sum of eigenvectors, and A is diagonalizable.

Thus

$$\boxed{A \text{ is diagonalizable} \iff m_A \text{ splits with no repeated root.}}$$

For the Jordan block above,

$$m_J(t) = (t-1)^2.$$

The repeated factor records the nilpotent motion along the generalized eigendirection.

§1.3 Projection as Best Approximation

An inner product adds lengths and angles to a vector space. For a subspace $W \subset \mathbb{R}^n$, the orthogonal projection of x onto W is the unique vector $p \in W$ for which

$$x - p \perp W.$$

If $Q \in \mathbb{R}^{n\times r}$ has orthonormal columns spanning W, then

$$p = QQ^T x.$$

Indeed, $QQ^T x \in W$, while

$$Q^T(x - QQ^T x) = Q^T x - Q^T x = 0.$$

The matrix

$$P_W = QQ^T$$

is symmetric and idempotent:

$$P_W^T = P_W, \qquad P_W^2 = P_W.$$

Orthogonality also proves optimality. For any $w \in W$,

$$x - w = (x - p) + (p - w)$$

is an orthogonal decomposition, so

$$\|x - w\|^2 = \|x - p\|^2 + \|p - w\|^2 \geq \|x - p\|^2.$$

Projection is therefore the closest point in the subspace. Least squares, Fourier approximation, and Gram–Schmidt all begin with this Pythagorean identity.

§1.4 Gram–Schmidt and QR

Given independent vectors $a_1, \dots, a_r$, define

$$u_1 = a_1,$$

$$u_j = a_j - \sum_{i<j} \frac{\langle a_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

Each u_j is orthogonal to the previous vectors, and independence guarantees $u_j \neq 0$. Normalizing gives orthonormal vectors

$$q_j = \frac{u_j}{\|u_j\|}.$$

For the columns

$$a_1 = (1,0,0,1)^T, \quad a_2 = (1,1,0,1)^T, \quad a_3 = (1,1,1,0)^T,$$

one obtains

$$q_1 = \frac{1}{\sqrt{2}}(1,0,0,1)^T, \qquad q_2 = (0,1,0,0)^T,$$

$$q_3 = \frac{1}{\sqrt{6}}(1,0,2,-1)^T.$$

Writing $A = (a_1, a_2, a_3)$ and $Q = (q_1, q_2, q_3)$, the coefficients

$$r_{ij} = \langle q_i, a_j \rangle$$

form the upper-triangular matrix

$$R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{6}/2 \end{pmatrix},$$

and

$$\boxed{A = QR, \qquad Q^T Q = I.}$$

QR factorization is Gram–Schmidt written as a matrix identity: Q stores the orthonormal geometry, while R stores how the original columns were assembled from it.

§1.5 Householder QR

Classical Gram–Schmidt can lose accuracy when columns are nearly dependent, because it subtracts nearly equal vectors. Modified Gram–Schmidt reduces the damage by updating residuals after each projection. Householder QR takes a different route: reflect an entire column onto a coordinate axis, with a crucial choice of sign that prevents a small subtraction.

For a nonzero real vector x, define

$$s = \begin{cases} 1, & x_1 \geq 0, \\ -1, & x_1 < 0, \end{cases} \qquad \alpha = -s\|x\|_2, \qquad v = x - \alpha e_1 = x + s\|x\|_2 e_1.$$

The sign convention at $x_1 = 0$ is intentional: the usual sign function with $\operatorname{sign}(0) = 0$ would make the proposed target $\alpha e_1 = 0$, which no orthogonal map can produce from

nonzero x. Moreover $|v_1| = |x_1| + \|x\|_2$, so the first component is formed by adding magnitudes rather than subtracting nearly equal numbers.

Now set

$$\boxed{H = I - 2\frac{vv^T}{v^Tv}.}$$

Because vv^T is symmetric and $(vv^T)^2 = (v^Tv)vv^T$,

$$H^T = H, \qquad H^2 = I.$$

It is therefore an orthogonal reflection. The choice $\alpha^2 = \|x\|_2^2$ also gives

$$v^Tv = 2(\|x\|_2^2 - \alpha x_1), \qquad 2v^Tx = v^Tv.$$

Consequently

$$Hx = x - \frac{2v(v^Tx)}{v^Tv} = x - v = \boxed{\alpha e_1}.$$

So the first column is sent exactly to the coordinate axis without repeatedly subtracting projections.

For a complete example, take

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Our stable sign choice is $\alpha = -5$ and $v = (8, 4)^T$. Thus

$$H = \begin{pmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{pmatrix}, \qquad HA = \begin{pmatrix} -5 & -4 \\ 0 & 3 \end{pmatrix} = R.$$

Since $H^T = H = H^{-1}$, putting $Q = H$ gives the exact QR factorization

$$\boxed{A = QR, \qquad Q^TQ = I.}$$

The upper-triangular factor need not have positive diagonal entries; if positive diagonals are preferred, multiplying a column of Q and the corresponding row of R by -1 does not change their product.

In exact arithmetic either target sign $\alpha = \pm\|x\|_2$ can work when $v \neq 0$; the *opposite* sign used here avoids cancellation in floating-point computation. This small choice helps explain why a numerically stable Householder implementation is more than the geometric formula for a reflection. The same mechanism underlies practical QR algorithms.[1]

§1.6 Orthogonal Maps and Reflections

A real matrix U is orthogonal when

$$U^TU = I.$$

It preserves inner products:

$$\langle Ux, Uy\rangle = x^TU^TUy = \langle x, y\rangle.$$

[1] For Householder QR and the noncancelling sign convention, see Trefethen and Bau, *Numerical Linear Algebra*, Lecture 10.

Hence it preserves lengths, angles, and orthogonality. Its determinant satisfies

$$(\det U)^2 = 1,$$

so

$$\det U = \pm 1.$$

The sign records orientation: rotations have determinant $+1$, while a single reflection has determinant -1.

Householder matrices show that reflections are not merely examples. Every orthogonal matrix is a product of reflections. QR factorization repeatedly uses this fact to rotate or reflect successive column spaces into coordinate position without changing their metric geometry.

Over $\mathbb{C}$, the corresponding maps are unitary:

$$U^*U = I.$$

Complex conjugation in the Hermitian inner product is essential; without it, $\langle x, x\rangle$ need not be positive.

§1.7 Spectral Theorem by Induction

Let $A = A^T$ be real symmetric. On the unit sphere, the Rayleigh quotient

$$R_A(x) = x^T Ax$$

attains a maximum because the sphere is compact. At a maximizing unit vector v, Lagrange multipliers give

$$2Av = 2\lambda v,$$

so v is an eigenvector.

Its orthogonal complement is invariant. If $w \perp v$, then

$$\langle Aw, v\rangle = \langle w, Av\rangle = \lambda\langle w, v\rangle = 0.$$

Thus A restricts to a symmetric operator on $v^\perp$. Repeating the argument inductively produces an orthonormal eigenbasis.

Therefore

$$\boxed{A = A^T \quad \Longrightarrow \quad A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)Q^T}$$

for an orthogonal matrix Q. The proof explains both parts of the theorem: symmetry makes the orthogonal complement invariant, while compactness supplies the first eigenvector.

The extreme Rayleigh values are the extreme eigenvalues:

$$\lambda_{\min}(A) = \min_{\|x\|=1} x^T Ax, \qquad \lambda_{\max}(A) = \max_{\|x\|=1} x^T Ax.$$

The spectral theorem is therefore also an optimization theorem.

§1.8 Spectral Projectors and Functions

Suppose a diagonalizable matrix has distinct eigenvalues $\lambda_1, \ldots, \lambda_s$. The Lagrange polynomials from the minimal-polynomial argument give projectors

$$P_i = \prod_{j \neq i} \frac{A - \lambda_j I}{\lambda_i - \lambda_j}.$$

They satisfy

$$P_i^2 = P_i, \qquad P_i P_j = 0 \ (i \neq j), \qquad \sum_i P_i = I,$$

and $P_i V = E_{\lambda_i}$. Hence

$$A = \sum_i \lambda_i P_i.$$

For any function f defined on the spectrum, set

$$f(A) = \sum_i f(\lambda_i) P_i.$$

This is finite-dimensional functional calculus.

If $A^2 = I$, the eigenvalues are ± 1, and

$$P_+ = \frac{I + A}{2}, \qquad P_- = \frac{I - A}{2}.$$

Therefore

$$\begin{aligned} e^{tA} &= e^t P_+ + e^{-t} P_- \\ &= \cosh t \, I + \sinh t \, A. \end{aligned}$$

A matrix exponential has been computed without multiplying matrices repeatedly. The projectors separate the invariant modes first, and the scalar function is then applied to each mode.

§1.9 Similarity versus Congruence

Similarity

$$A \longmapsto P^{-1} A P$$

changes the basis of a linear operator and preserves eigenvalues. Congruence

$$A \longmapsto P^T A P$$

changes variables in a quadratic form

$$q(x) = x^T A x, \qquad x = Py.$$

It need not preserve eigenvalues.

Consider

$$q(x, y) = 2x^2 + 4xy - y^2.$$

Completing the square gives

$$q(x, y) = 2(x + y)^2 - 3y^2.$$

With

$$u = x + y, \qquad v = y,$$

the form becomes

$$q = 2u^2 - 3v^2.$$

Thus it has one positive and one negative square. Sylvester's law of inertia says that the triple

$$(n_+, n_-, n_0)$$

of positive, negative, and zero squares is invariant under every invertible congruence. Here the inertia is

$$(1, 1, 0).$$

Orthogonal diagonalization is a special congruence that also happens to be a similarity for symmetric matrices. It preserves Euclidean geometry as well as the quadratic form's spectral data; a general completing-square substitution preserves only the inertia classification.

§1.10 Tests for Positive Definiteness

For a symmetric matrix A, the following are equivalent:

$$x^T Ax > 0 \quad (x \neq 0),$$

$$\lambda_i(A) > 0 \quad \text{for every } i,$$

$$\Delta_1 > 0, \ldots, \Delta_n > 0,$$

where Δ_k are the leading principal minors, and

$$A = LL^T$$

for an invertible lower-triangular matrix L with positive diagonal. These are the quadratic-form, spectral, Sylvester, and Cholesky criteria.

For

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

the leading principal minors are

$$\Delta_1 = 2, \qquad \Delta_2 = 3, \qquad \Delta_3 = 4.$$

Hence $A \succ 0$. Cholesky elimination gives

$$L = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & \sqrt{3/2} & 0 \\ 0 & -\sqrt{2/3} & 2/\sqrt{3} \end{pmatrix},$$

and a direct multiplication verifies

$$A = LL^T.$$

The factorization makes positivity immediate:

$$x^T Ax = \|L^T x\|^2 > 0$$

for $x \neq 0$.

§1.11 Principal axes organize conics, Hessians, and PCA

A quadratic polynomial in two variables can be written

$$F(x) = x^T Ax + b^T x + c, \qquad A = A^T.$$

An orthogonal change aligns the coordinates with eigenvectors of A, removing cross terms. A translation then completes squares and moves the center when one exists. Rotation and translation perform different jobs: the first changes directions, the second changes the origin.

The same matrix appears as a Hessian in optimization. Near a critical point p,

$$f(p+h) = f(p) + \frac{1}{2} h^T \nabla^2 f(p) h + o(\|h\|^2).$$

The inertia of the Hessian distinguishes a local minimum, maximum, or saddle when the Hessian is nonsingular.

PCA applies the same spectral geometry to data. Consider the centered points

$$(2,1),\ (1,1),\ (-1,-1),\ (-2,-1).$$

Their covariance matrix is

$$C = \frac{1}{4} \sum_{i=1}^{4} x_i x_i^T = \begin{pmatrix} 5/2 & 3/2 \\ 3/2 & 1 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{7 \pm 3\sqrt{5}}{4}.$$

For λ_+, an eigenvector can be chosen proportional to

$$\begin{pmatrix} 1 \\ (\sqrt{5}-1)/2 \end{pmatrix}.$$

Projection onto this direction preserves the largest possible average squared variation. PCA is therefore the Rayleigh-quotient extremum of the covariance quadratic form, not a separate algorithmic miracle.

§1.12 Projection in Hilbert Spaces

In a Hilbert space H, every closed subspace M still has an orthogonal projection. To see why closedness matters, take a minimizing sequence $m_n \in M$ for the distance from x to M. The parallelogram identity shows that (m_n) is Cauchy; completeness of M gives a limit $m \in M$, and the usual first-variation argument gives

$$x - m \perp M.$$

Thus

$$H = M \oplus M^\perp.$$

Finite-dimensional least squares is the first instance of this projection theorem.

The spectral theorem also survives, but not always as an eigenbasis. On

$$H = L^2([0,1]),$$

consider the self-adjoint multiplication operator

$$(Mf)(t) = tf(t).$$

If $Mf = \lambda f$, then

$$(t - \lambda)f(t) = 0$$

almost everywhere. The set $\{t = \lambda\}$ has measure zero, so $f = 0$ in L^2. The operator has no nonzero eigenvectors, even though its spectrum is the whole interval $[0, 1]$.

The correct infinite-dimensional statement uses a projection-valued measure:

$$M = \int_{[0,1]} \lambda \, dE(\lambda).$$

Finite diagonal matrices are the discrete case of this integral. Orthogonal projection, self-adjointness, and functional calculus remain; what can disappear is a basis made from individual eigenvectors.